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### Antifuse device having interconnect jumper

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#### Abstract

An antifuse device, including a gate having a gate dielectric layer; a first doping region connected to a first end of the gate; a second doping region connected to a second end of the gate, the second end being opposite to the first end of the gate; a channel that is disposed under the gate and that connects the first doping region and the second doping region; and an interconnection jumper that electrically connects the first doping region and the second doping region.

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## Background/Summary

### TECHNICAL FIELD

(1) The present disclosure generally relates to semiconductor devices, and more particularly relates to semiconductor antifuse device having interconnect jumper.

### BACKGROUND

(2) Microelectronic devices, such as memory devices, microprocessors, typically include electrical antifuse devices configured to program ingredients of the memory device or provide access to redundant circuitries. In general, antifuse device can be operated by dielectric layer breakdown upon applying an electric field across two electrodes located on opposite sides of the dielectric layer to induce a rupture, therefore causing reduction of the dielectric layer across the two electrodes. The antifuse device can be implemented using Metal-Oxide-Semiconductor Field Effect Transistor (MOSFET) process to achieve a short state through its gate dielectric layer. Moreover, conventional memory product includes planar MOSFET antifuse device that relies on a source side gate overhang. One of the challenges in implementing electrical antifuse device is a required high success rate of gate dielectric layer breakdown. In addition, with a trend of scaling the dimension of semiconductor memory devices, there is a need to further reduce the electrical antifuse device area.

## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A depicts a circuit diagram of an antifuse device having interconnect jumper according to embodiments of the present technology.

- (2) FIG. 1B depicts a layout of an antifuse device having interconnect jumper according to embodiments of the present technology.
- (3) FIG. 2 depicts a cross-sectional schematic diagram of an antifuse device having interconnect jumper according to embodiments of the present technology.
- (4) FIGS. 3A and 3B depict a FinFET device having a protruded fin with enhanced electric fields.
- (5) FIG. 4A depicts a circuit diagram of an antifuse device according to embodiments of the present technology.
- (6) FIG. 4B depicts a layout of an antifuse device according to embodiments of the present technology.
- (7) FIG. 5A depicts a circuit diagram of an antifuse device according to embodiments of the present technology.
- (8) FIG. 5B depicts a layout of an antifuse device according to embodiments of the present technology.
- (9) FIGS. 6A and 6B depict cross-sectional schematic diagrams of antifuse devices with and without interconnect jumper respectively, according to embodiments of the present technology.
- (10) FIG. 7 is a flow chart illustrating a method of fabricating an antifuse device with an interconnect jumper according to embodiments of the present technology.
- (11) FIG. 8 is a schematic view of a system that includes a semiconductor device configured according to embodiments of the presented technology.
- (12) The drawings illustrate only example embodiments and are therefore not to be considered limiting in scope. The elements and features shown in the drawings are not necessarily to scale, emphasis instead being placed upon clearly illustrating the principles of the example embodiments. Additionally, certain dimensions or placements may be exaggerated to help visually convey such principles. In the drawings, the same reference numerals used in different embodiments designate like or corresponding, but not necessarily identical, elements.

#### DETAILED DESCRIPTION

- (13) Conventional MOSFET antifuse devices are not suitable to high density memory cell applications. They have limitations in redundancy and error correction due to variability of the fuse resistance and increase in fuse resistance along with a trend of device scaling. For example, the fuse resistance of the conventional MOSFET antifuse device increases when the gate width decreases with a given power supply applied thereon. Specifically, conventional planar MOSFET antifuse devices rely on only a source side gate overhang and gate dielectric breakdown therein to change the antifuse device from an open gate-source connection to a shorted gate-source connection.
- (14) To address these drawbacks and others, embodiments of the present technology provide antifuse devices using planar MOSFET or FinFET processes so as to achieve an improved fuse success rate and scalable antifuse device area. The antifuse devices can be implemented using the gate dielectric layer made of standard MOSFET process to exploit gate dielectric break down to short the antifuse device, and thereby does not need an additional mask step or process for the antifuse device implementation in memory products. Specifically, the antifuse devices utilize both a source node and a drain node of the MOSFET for the antifuse functions. Further, an interconnect jumper is implemented into the antifuse devices, electrically connecting the source node and the drain node of the antifuse devices. When programming, a voltage level of the drain node can be configured to be same to the source node, e.g., at 0V, through the interconnect jumper. By utilizing both of the source and drain nodes of the FET, the described antifuse devices can form primary breakdown regions at the source end as well as the node end of the channel, therefore creating two breakdown paths, i.e., a gate-source path and a gate-drain path. The enlarged primary breakdown regions enhance the chance of gate dielectric breakdown and improve the fuse success rate of the antifuse devices. Moreover, the width of the antifuse devices and an overall antifuse array area can be reduced without degrading the fuse success rate.

(15) FIG. 1A shows an antifuse device **100** that is connected to a selector transistor **110**. In this embodiment, the antifuse device **100** is implemented using planar MOSFET process to achieve a short state through its gate oxide layer. The antifuse device **100** can be constructed according on a structure of field effect transistor (FET). For example, the antifuse device **100** includes a gate **102**, a source node **104**, and a drain node **106**. Specifically, the antifuse device **100** includes an interconnect jumper **108** that electrically connects the source node **104** and the drain node **106**. In this example, the selector transistor **110** is connected to the antifuse device **100** by sharing a drain node of the selector transistor **119** with the source node **104** of the antifuse device **100**. The selector transistor **110** can also be implemented by using planar MOSFET process and includes a gate **112** and a source node **114**.

(16) The gate **102** of the antifuse device **100** may be connected to a metal gate wordline (WL) of a memory device and the gate **112** of the selector transistor **110** may be connected to a select voltage which is configured to control/activate the antifuse device **100**. When programming, the source node **114** of the selector transistor **120** may be connected to ground and the select voltage, e.g., 1V, may be applied on the gate **112**. The selector transistor **110** is then turned on to ground the shared source node **104** of the antifuse device **100** to close to 0V. Here, the drain node **106** is also turned close to 0V because its voltage level is leveraged similarly to the source **104** through the electrical connection of the interconnect jumper **108**. In addition, the memory WL may supply a high voltage, e.g., 5V, to the gate **102** of the antifuse device **100** in order to form a high  $V_{sub,gs}$  and  $V_{sub,gd}$  across the source node **104** and the drain node **106**, respectively.

(17) In some embodiments, similar to conventional planar MOSFET device, the gate **102** of the antifuse device **100** includes a gate dielectric layer disposed on a channel that connects the source node **104** and the drain node **106**. In particular, the gate **102** overhangs at least a portion of the source node **104** and a portion of the drain node **106**. When programming the antifuse device **100** with a high gate voltage, the gate dielectric layer disposed close to the gate overhang regions may break down and lead to ohmic-like and symmetric post break down characteristics. The rupture of the gate dielectric layer form one or more conductive channels from the gate **102** to the source node **104** or the drain node **106**, as arrows marked in FIG. 1A.

(18) FIG. 1B shows a layout of the antifuse device **100** that includes the interconnect jumper **108** and that is connected with the selector transistor **110**. This layout is presented in a plan view, including the gate **102**, the source node **104**, the drain node **106**, and the interconnect jumper **108** of the antifuse device **100**. The source node **104** and the drain node **106** are vertically aligned and a channel region is disposed therebetween. The gate **102** of the antifuse device **100** is aligned horizontally above the channel region. Further, the gate dielectric layer is disposed on a top surface of the channel region and below the metal gate. This layout also includes the selector transistor **110** that is connected to the antifuse device **100** through a shared source **104**. Here, the selector **120** includes the gate **112** that is aligned in parallel to the gate **102** of the antifuse device **100**.

(19) In some embodiments and as shown in FIG. 1B, the gate **102** overhangs a portion of the source node **104** and a portion of the drain **106**. When programming, the magnitude of local electrical field is enhanced in the gate overhang regions, causing inverted channel regions **130** that are align along the gate width direction. In this example, the antifuse device **100** includes two inverted channel regions **130** that are disposed close to the source node **104** and the drain node **106**, respectively. These two inverted channel regions **130** may not be merged.

(20) As shown in FIG. 1B, the interconnect jumper **108** can be disposed above the gate **102** of the antifuse device **100**. It can be connected to the source node **104** and the drain node **106** though the source and drain contacts. The interconnect jumper **108** may have a length that similar or smaller than the length of the MOSFET implemented into the antifuse device **100**. Further, the interconnect jumper **108** may have a width equal to or smaller than the gate width of the antifuse device **100**. In addition, the interconnect jumper **108** may be made of at least one of copper, tungsten, molybdenum, nickel, titanium, tantalum, platinum, silver, gold, ruthenium, iridium, rhenium,

rhodium, or alloys thereof.

(21) FIG. 2 shows a cross-sectional schematic diagram of the antifuse device **100** implemented from a conventional MOSFET. The antifuse device **100** comprises the gate **102** disposed on a gate dielectric layer **150**, the source **104**, the drain **106**, and the substrate **140**. In some embodiments, the gate **102** may be formed of polysilicon or conductive metals. The gate dielectric layer **150** may be made of gate oxide, silicon dioxide, and/or silicon nitride. In some other embodiments, the gate dielectric layer **150** may comprise two gate dielectric portions, e.g., a thicker high-K dielectric layer and a thinner interfacial layer. The source and drain regions **104** and **106** are formed within the substrate **140** and may be heavily doped. In addition, the antifuse device **100** includes a depletion region **132**. Generally, the source region **104** and the drain region **106** have higher concentration of dopants than the rest of the substrate **140**. The source and drain regions **104** and **106** can be disposed within the depletion region **132**. Further, the region in the substrate **140** is directly below the gate **102** and between the source region **104** and the drain region **106** is referred as a channel **142**. In this example, the antifuse device **100** also includes the interconnect jumper **108** that is disposed above the gate **102** and electrically connects the source region **104** and the drain region **106**. The interconnect jumper **108** may be isolated from other ingredients of the MOSFET by one or more dielectric materials.

(22) In an exemplary embodiment, the antifuse device **100** may be fabricated based on NMOS transistor processes, and the source and drain regions **104** and **106** may contain a higher concentration of n-type dopants. Moreover, the depletion region **132** can be an insulating region within a conductive/doped semiconductor material, e.g., the substrate **140**, where the mobile charge carriers have been diffused away or have been forced away by an electric field. In this example, the depletion region **132** may be formed due to the electron carriers being drawn to the inverted channel region **130**. As shown in FIG. 2, the gate **102** as well as gate dielectric layer **150** overhang a portion of the source region **104** and a portion of the drain region **106**. In other words, the source and drain regions **104** and **106** extend toward the channel and under the gate **102**. The gate dielectric breakdown paths are illustrated in FIG. 2 in arrows. The breakdown of the gate dielectric layer **150** can be caused by applying a high voltage  $V_g$  at the gate **102** while the source and drain **104** and **106** are grounded.

(23) The antifuse device **100** is normally in an open status until a programming voltage is applied on its gate **102**. By applying the gate voltage, a resulting current will lead the antifuse device **100** to permanently break down or blow the gate dielectric layer **150** to rupture. Once the antifuse device **100** is break down, it acts as a closed short circuit. Generally, the gate dielectric layer **150** is ruptured at the gate overhang regions where the source and drain doping levels are high and a low resistance path can be formed between the gate to the source **104** or the drain **106**. The carrier concentration in the channel is lower than the source **102** or the drain **104**, therefore a fully converted region can be formed under the gate overhang regions where a higher electric field exists when the programming voltage is applied to the gate **102**. In particular, the antifuse device **100** may include two primary breakdown regions **144** that are close to the gate overhang regions and that correspond to gate dielectric break down paths in the antifuse device **100**.

(24) As discussed, the antifuse device **100** includes the drain node **106**, which has a voltage level similar to the source node **104**. The leverage of source **104** and drain **106** voltages are achieved by the electrical interconnection through the interconnect jumper **108**. Compare to traditional planar MOSFET implemented antifuse device which does not include an active drain, the antifuse device **100** utilizes both of the source **104** and the drain **106** for gate dielectric breakdown. The antifuse device **100** contains two inverted channel regions **130** for blowing the gate dielectric layer **150**, doubling a total perimeter of the antifuse device **100** that is vulnerable to the gate dielectric breakdown. In some embodiments, a planar MOSFET with reduced gate width can be implemented in the present technology for the antifuse device scaling. As discussed, the antifuse device **100** includes doubled inverted channel regions **130** compared to traditional applications, therefore the

implemented MOSFET with reduced gate width can achieve a similar or greater success rate on gate dielectric breakdown. The width reduced MOSFET can be introduced to the present technology to reduce the antifuse device width and improve overall fuse array area density in the memory device.

(25) FIGS. 3A and 3B illustrate a FinFET device **300** having enhanced electric fields at the fin top corners. For example, FIG. 3A shows a cross sectional view of the FinFET device **300** which includes a fin **302** protruding from a substrate **304**. The fin **302** can be patterned from an epitaxial layer deposited on the substrate **304** or etched out of the substrate **304**. The fin **302** can be made of materials including silicon and silicon germanium. A bottom portion of the fin **320** is surrounded by a shallow trench isolation (STI) for electrical isolation from the FinFET device **300** to the substrate or adjacent devices disposed on the substrate **304**. In addition, an exposed portion of the fin **302** is encapsulated by the gate **306** through a gate dielectric layer **306**. The gate **306** may be made of metal gate materials including titanium nitride, tungsten nitride, tantalum nitride, and/or molybdenum nitride. The gate dielectric **308**, as shown in FIG. 3B, may include a high-K dielectric layer **312** and an interfacial layer **314**. The high-k dielectric layer can be made of  $\text{HfO}_2$  and the interfacial layer can be  $\text{SiO}_2$ . Generally, the interfacial layer **314** and the high-k dielectric layer **314** are sequentially formed on the exposed fin **302**. The FinFET device **300** also includes a source region and a drain region (not shown) that are disposed at both ends of the fin **302**.

(26) As shown, the FinFET device **300** may be a n-channel FET in which a positive gate to source voltage is needed to create a conductive channel between the source and drain. The positive gate voltage attracts free floating electrons within the fin body towards the top surface and sidewalls of the fin **302**, forming the n type channel. In some other examples, the FinFET device **300** may be a p-channel FET in which a negative gate to source voltage is needed to create a conductive channel between the source and drain. The negative gate voltage attracts free floating holes within the fin body towards the top surface and sidewalls of the fin, forming the p type channel.

(27) Because of the fin geometrical shape, there is a charge injection at fin corners which is in proportional to the curvature radius at fin corners. For FinFET device **300**, the electric field at its fin top corners is always amplified compared to the electric field at its sidewall. For example, the electric field at fin top corners can be as high as  $2 \times 10^7$  V/cm and the electric field at fin sidewall can be reduced to around  $1 \times 10^7$  V/cm when a same gate voltage is applied. As a result, the fin corner effect improves the performance of the FinFET device **300** through enhancing its on-current at the fin corners. Meanwhile, the enhanced electrical field may cause an easier gate dielectric breakdown at the fin top corners compare to that on the fin top surface and fin sidewalls. In addition, the interfacial layer **314** disposed at fin bottom edge usually has a relatively poor film quality compared to the interfacial layer **314** disposed on fin sidewall and top surface, due to process limitations related to the geometry restraints thereon. Therefore, the fin bottom edge region may be a secondary gate dielectric breakdown region in which the interfacial layer **314** ruptures.

(28) FIG. 4A shows an antifuse device **400** connected with a selector transistor **410**. Both of the antifuse device **400** and the selector transistor **410** can be implemented using FinFET processes. Specifically, the antifuse device **400** can be fabricated based on FinFET device structure. In some embodiments, the antifuse device **400** includes a gate **402**, a source node **404**, and a drain node **406**. The selector transistor **410** can also be fabricated based on the FinFET device structure and includes a gate **412**, a source **414**, and a drain that is shared with the source **404** of the antifuse device **400**. The selector transistor **410** is connected to the antifuse device **400** at the source **404**.

(29) In some embodiments, the gate **402** of the antifuse device **400** may be connected to a metal gate WL of a memory device and the gate **412** of the selector transistor **410** may be connected to a select voltage configured to control/select the antifuse device **400**. When programming, the source node **414** of the selector transistor **410** is grounded and a select voltage, e.g., 1V, can be applied to the gate **412**. The selector transistor **410** is then turned on to ground the shared source node **404** of the antifuse device **400** to close to 0V. In this example, the drain node **406** of the antifuse device

**400** may be floating and present a high impedance. The drain node **406** is included here in the antifuse device **400** in order to comply with the FinFET device design rule, e.g., a constant contacted poly pitch (CPP) throughout a die. Once the source node **404** of the antifuse device **400** is configured to be close to ground, the memory WL may supply a high voltage, e.g., 5V, to the gate **402** to form a high  $V_{sub,gs}$  across the gate **402** and the source **404**.

(30) As discussed in FIG. 3, FinFET device includes a gate dielectric layer wrapping up the exposed fin. In this example, the antifuse device **400** includes a gate dielectric layer encapsulating a top surface and sidewalls of the fin which connects the source node **404** and the drain node **406**. When programming the antifuse device **400** with a high gate voltage, the gate dielectric layer may break down along a path from the gate **402** to the source **404**, indicating by the arrow. The blowing of the gate dielectric may happen at the fin top corners, where the electric field is amplified due to the fin corner geometry. In addition, the gate dielectric layer of the antifuse device **400** may rupture at the fin bottom edge, where an interfacial layer of the gate dielectric layer has a relatively poor quality due to its process limitations related to the geometry restraints at fin bottom.

(31) Turning to FIG. 4B, which shows a layout of the antifuse device **400** and the selector transistor **410**. This layout is presented in a plan view, including the gate **402**, the source node **404**, and the drain node **406** of the antifuse device **400**. The source node **404** and the drain node **406** are vertically aligned and each connects to one end of the fin **408**. The gate **402** of the antifuse device **400** is aligned perpendicular to the fin **408** length direction. In addition, the gate **402** includes a gate dielectric layer that is disposed on a top surface and two sidewalls of the fin **408**. This layout also shows the selector transistor **410** connected to the antifuse device **400** through the shared source **404**. Specifically, the selector transistor **410** includes the gate **412** that is aligned in parallel to the gate **402** of the antifuse device **400**.

(32) In some embodiments, the antifuse device **400** may include two inverted channel regions **420** as marked in FIG. 4B. The inverted channel regions **420** may be disposed at the top corners of the fin **408** and extend from the source side edge of the fin toward the drain node **406**. The length and width of each of the inverted channel region **420** may be in proportional to the programming voltage applied on the gate **402**. When programming, the antifuse device **400** may form enhanced electrical fields and the inverted channel regions **420** at the fin top corners, causing the gate dielectric to permanently rupture therethrough. One or more electrically short paths may be formed from the gate **402** to the source **404** of the antifuse device **400**, as shown by the arrows.

(33) In some other embodiments, the antifuse device **400** may form one or more electrical short paths through the fin **408** bottom edges, where the interfacial layer of the gate dielectric has a relatively poor quality. When programming, the interfacial layer at the fin **408** bottom edge may permanently rupture and form electrical short paths therethrough.

(34) FIGS. 5A and 5B show another antifuse device **500** that is connected to a selector **510**. Similar to the antifuse device **400** and the selector **410**, and as shown in FIG. 5A, the antifuse device **500** and the selector **510** can be implemented using FinFET processes. For example, the antifuse device **500** may include a gate **502**, a source node **504**, and a drain node **506**. The selector transistor **510** may include a gate **512**, a source **514**, and a drain that is shared with the source **504** of the antifuse device **500**. The selector transistor **510** is connected to the antifuse device **500** at the source **504**. In this example, the antifuse device **500** also includes an interconnect jumper **508** that electrically connects the source node **504** and the drain node **506**.

(35) In some embodiments, the gate **502** of the antifuse device **500** may be connected to a metal gate WL of a memory device and the gate **512** of the selector transistor **510** may be connected to a select voltage configured to control/select the antifuse device **500**. When programming, the source node **514** of the selector transistor **510** is grounded and a select voltage, e.g., 1V, can be applied to the gate **512**. The selector transistor **510** is then turned on to ground the shared source node **504** of the antifuse device **500** close to 0V. The drain node **506** of the antifuse device **500** will also have a voltage level close to 0V as it is electrically connected to the source node **504** through the

interconnect jumper **508**. In this example, the memory WL may apply a high voltage, e.g., 5V, on the gate **502** to program the antifuse device **500** by forming a high V<sub>gs</sub> and V<sub>gd</sub> across the source node **504** and the drain node **506**, respectively. A gate dielectric layer of the antifuse device **500** may break down and lead to one or more short paths from the gate **502** to the source node **504** or the drain node **506**, permanently converting the antifuse device **502** to a short state.

(36) Turning now to FIG. 5B, which shows a layout of the antifuse device **500** and the selector transistor **510**. The plan view layout shows that the antifuse device **500** includes the gate **402**, the source node **404**, the drain node **406**, and the interconnect jumper **508**. In some embodiments, the interconnect jumper **508** can be disposed above the fin **508** and the gate **502**. In addition, the interconnect jumper **508** is connected with the source node **504** and the drain node **506** to equate the voltage levels therebetween. In this example, the interconnect jumper **508** is wider than the fin **508** along the horizontal direction. In some other embodiments, the interconnect jumper **508** may have a width equal to or less than that of the fin **508**.

(37) In some embodiments and when programming, the antifuse device **500** may include inverted channel regions **520** that are disposed on fin top corners and along the gate length. Specifically, the inverted channel regions **520** may extend across the whole gate length. As shown in FIG. 5A, the antifuse device may have two short paths from the gate **502** to the source node **504** or the drain node **506**, respectively. When applying a programming voltage to the gate **502**, inverted channel regions **520** within the fin **508** may be formed at source side fin edge corners and the drain side fin edge corners, extending to the drain side and the source side respectively along the fin top corners. Under certain high gate voltage, e.g., 5V, the inverted channel regions **520** originated from both ends of fin top corners may merge at a middle region of the fin **508**, showing continuous fully inverted regions as shown in FIG. 5B.

(38) Compared to the inverted channel regions **420** of the antifuse device **400**, the inverted channel regions **520** of the antifuse device **500** may substantially increase the length of the inverted channel regions when programming. Since the FinFET based antifuse device gate dielectric breakdown is correlated to the area of fully inverted region, therefore the elongated inverted channel regions **520** enhance the fuse success rate when operating the antifuse device **500**.

(39) FIGS. 6A and 6B show cross sectional views of schematic diagrams of antifuse device **400** and **500**, respectively. Both diagrams are taken across a plane of the fin sidewall. As discussed, the antifuse devices **400** and **500** are implemented using FinFET processes. For example, the antifuse device **400** includes a channel **602** of part of the fin **408** disposed on substrate **410**, the source **404** and the floating drain **406** being connected to both ends of the channel **602**. The gate **402** has an orientation at right angles to the fin **408** and surrounds the top surface and sidewalls of the fin **408**. In particular, the gate **402** includes a gate dielectric layer **422** directly disposed on the top surface and sidewalls of the fin **408**.

(40) In this example, the fin **408** provides the channel **602** between the source **404** and the drain **406**. The gate **402** is configured to surround the channel from three sides of the fin **408** in order to provide electrostatic control of the channel with fully depleted region of carriers. When programming, the source node **404** is grounded and the drain node **406** keep floating. Upon application of electric field across the gate dielectric layer **422**, the magnitude of the electrical field can be locally enhanced at the boundary between a high-K dielectric portion and an interfacial dielectric portion of the gate dielectric layer **422**, due to the Fin **408** geometry. The voltage applied on gate **402** may form the inverted channel regions **420** in the channel at each of the fin **408** top corners, where enhanced electric fields exist. Specifically, the inverted channel regions **420** extend toward the floating drain node **406**, each having a width W<sub>1</sub> and a length L<sub>1</sub>, proportional to the applied gate voltage. As shown in FIG. 6A, applying gate voltage and tying the source node **404** to ground can not create a fully inverted channel between the source and drain, because the drain node **406** is floating and no depletion region exists therein. In this example, programming the antifuse device **400** may cause the gate dielectric breakdown at the top corners of the fin **408**, where the



inverted channel regions **420** are formed.

(41) The antifuse device **500** is implemented by FinFET process similar to that of the antifuse device. For example, it includes the channel **604** disposed on substrate **510**, the source **504** and the drain **506** being connected to both ends of the channel **604**. The gate **502** has an orientation at right angles to the fin **508** and surrounds the top surface and sidewalls of the fin **508**. The gate **502** includes gate dielectric layer **522** directly disposed on the top surface and sidewalls of the fin **508**. The antifuse device **500** also includes the interconnect jumper **508** that electrically connects the source **504** and the drain **506**. As shown in FIG. **6B**, a horizontal portion of the interconnect jumper **508** may be disposed above the FinFET structure implemented. Electrical contacts may be formed between the interconnect jumper **508** to the source **504** and the drain **506**. In addition, the interconnect jumper **508** may be made of at least one of copper, tungsten, molybdenum, nickel, titanium, tantalum, platinum, silver, gold, ruthenium, iridium, rhenium, rhodium, or alloys thereof.

(42) In this example, the source node **504** and the drain node **506** have a same voltage level, leveraged by the interconnect jumper **508**. When programming, the source node **504** is grounded through the activation of the selector transistor **510**. As a result, the drain node **506** is also grounded through the electrical interconnection by the jumper **508**. Here, depletion regions are formed in the source node region **504** as well as the drain node region **506**. As shown in FIG. **6B**, applying a gate voltage on the gate **502** may form two inverted channel regions **520** in the channel direction at each of the fin **508** top corners, where enhanced electric fields exist. The inverted channel regions **520** may locate at both ends of the channel and each extends toward the opposite fin end along the fin **508** top corners. The inverted channel regions **520** may be merged and have a length  $L_2$ . In this example, programming the antifuse device **500** may cause the gate dielectric breakdown at the top corners of the fin **508**, where two corresponding breakdown paths can be formed.

(43) As shown in FIGS. **6A** and **6B**, the antifuse device **500** may include inverted channel regions **520** having a doubled length to that of the antifuse device **400**. A fuse success rate of the antifuse device **500** may be also higher than that of the antifuse device **400**, because the fuse success rate is determined by the chance of gate dielectric break down which is further proportional to the area of fully inverted channel regions. Here, utilizing the existing drain node **506** of the FinFET structure, the antifuse device **500** leverages the voltage levels of the source node **504** and drain node **506** through the interconnect jumper **508**. This design also provides two potential breakdown paths, i.e., gate-source and gate-drain, and additional primary breakdown regions when programming the antifuse device **500**, therefore enhancing its fuse success rate.

(44) In this example, the source regions and drain regions can be formed through ion implantation in corresponding regions of the fin, doping various types of semiconductor materials to the source regions and drain regions, respectively. In some other examples, the source regions and drain regions of the FinFET antifuse device can be formed through etching of the fin to form trenches and then regrowing semiconductor materials such as SiGe therein.

(45) Turning now to FIG. **7**, a flow chart illustrating a method **700** of forming an antifuse device by implementing MOSFET processes. The method **700** includes providing a MOSFET structure, at **702**. For example, fabricating the antifuse device **100** by impending planar MOSFET processes to form the transistor device structure shown in FIGS. **1A**, **1B**, and **2**. In this example, the planar MOSFET device may include the source node **104**, the drain node **106** and a channel **142** disposed there between. The planar MOSFET device may also include a gate including gate dielectric layer **150** disposed above the channel **142** and overhangs a portion of each of the source node **104** and the drain node **106**. In another example, the antifuse device **500** can be fabricated by implementing FinFET processes. For example, the antifuse device **500** may include the source node **504**, the drain node **506**, and the fin **508**, wherein the source and drain nodes **504** and **506** are connected by the fin **508**. The antifuse device **500** also includes the gate including the gate dielectric layer **522** disposed on the top surface and sidewalls of the fin **508**.

(46) The method **700** also includes coating a passivation layer above the MOSFET structure, at

**704.** For example, a dielectric layer may be coated on the MOSFET device. Specifically, the dielectric layer may cover a gate region, a source region, and a drain region of the MOSFET structure. In one example, the passivation layer may be coated above the source region **104**, the drain region **106**, and the gate **102** of the antifuse device **100**. In another example, the passivation layer may be coated above the source region **504**, the drain region **506**, and the gate **502** of the antifuse device **500**. The dielectric layer may be made of tetraethyl orthosilicate (TEOS), silicon oxide (SiO), silicon nitride (SiN), silicon borocarbonitride (SiBCN), silicon oxycarbonitride (SiOCN), silicon oxycarbide (SiOC), silicon carbonitride (SiCN), silicon boronitride (SiBN), a low-k dielectric material, or a combination thereof.

(47) Further, the method **700** includes patterning the passivation layer to form vias above the source and drain regions of the MOSFET and a trench that is disposed on a top portion of the passivation layer and that is above the gate of the MOSFET, at **706**. For example, the dielectric layer may be patterned above the source and drain regions **104** and **106**, and fill with conductive materials therein to form via contacts. In addition, a top portion of the dielectric layer disposed above the gate **102** of the antifuse device **100** may be patterned to form a trench connected to the vias. In another example, the dielectric layer may be patterned above the source and drain regions **504** and **506**, and fill with conductive materials therein to form via contacts. In addition, a top portion of the dielectric layer disposed above the gate **502** of the antifuse device **500** may be patterned to form a trench connected to the vias.

(48) Lastly, the method **700** includes coating conductive materials in the patterned trench of the passivation layer to complete an interconnect jump that electrically connects the source and drain regions, at **708**. For example, conductive materials can be coated above the gate **102** of the antifuse device **100** and connected with the vias to electrically connects the source and drain **104** and **106**. In another example, conductive material can be coated above the gate **502** of the antifuse device **500** and connected with the vias to electrically connects the source and drain **504** and **506**. The conductive materials can be made of at least one of copper, tungsten, molybdenum, nickel, titanium, tantalum, platinum, silver, gold, ruthenium, iridium, rhenium, rhodium, or alloys thereof. Moreover, the conductive materials can be deposited into the patterned top portion of the gate by using chemical vapor deposition, physical vapor deposition, atomic layer deposition, plating, electroless plating, spin coating, and/or other suitable techniques.

(49) Any one of the semiconductor structures described above with reference to FIGS. **1-6** can be incorporated into any of a myriad of larger and/or more complex systems, a representative example of which is system **800** shown schematically in FIG. **8**. The system **800** can include a semiconductor device **810**, a power source **820**, a driver **830**, a processor **840**, and/or other subsystems or components **850**. The semiconductor device **810** can include features generally similar to those of the semiconductor devices described above, and can therefore include MOSFET structures for antifuse functions. The resulting system **800** can perform any of a wide variety of functions, such as memory storage, data processing, and/or other suitable functions. Accordingly, representative systems **800** can include, without limitation, hand-held devices (e.g., mobile phones, tablets, digital readers, and digital audio players), computers, and appliances. Components of the system **600** may be housed in a single unit or distributed over multiple, interconnected units (e.g., through a communications network). The components of the system **800** can also include remote devices and any of a wide variety of computer-readable media.

(50) Specific details of several embodiments of semiconductor devices, and associated systems and methods, are described below. A person skilled in the relevant art will recognize that suitable stages of the methods described herein can be performed at the wafer level or at the die level. Therefore, depending upon the context in which it is used, the term “substrate” can refer to a wafer-level substrate or to a singulated, die-level substrate. Furthermore, unless the context indicates otherwise, structures disclosed herein can be formed using conventional semiconductor-manufacturing techniques. Materials can be deposited, for example, using chemical vapor deposition, physical

vapor deposition, atomic layer deposition, plating, electroless plating, spin coating, and/or other suitable techniques. Similarly, materials can be removed, for example, using plasma etching, wet etching, chemical-mechanical planarization, or other suitable techniques.

(51) In accordance with one aspect of the present disclosure, the semiconductor devices illustrated above could be memory dice, such as dynamic random access memory (DRAM) dice, NOT-AND (NAND) memory dice, NOT-OR (NOR) memory dice, magnetic random access memory (MRAM) dice, phase change memory (PCM) dice, ferroelectric random access memory (FeRAM) dice, static random access memory (SRAM) dice, or the like. In an embodiment in which multiple dice are provided in a single assembly, the semiconductor devices could be memory dice of a same kind (e.g., both NAND, both DRAM, etc.) or memory dice of different kinds (e.g., one DRAM and one NAND, etc.). In accordance with another aspect of the present disclosure, the semiconductor dice of the assemblies illustrated and described above could be logic dice (e.g., controller dice, processor dice, etc.), or a mix of logic and memory dice (e.g., a memory controller die and a memory die controlled thereby).

(52) The devices discussed herein, including a memory device, may be formed on a semiconductor substrate or die, such as silicon, germanium, silicon-germanium alloy, gallium arsenide, gallium nitride, etc. In some cases, the substrate is a semiconductor wafer. In other cases, the substrate may be a silicon-on-insulator (SOI) substrate, such as silicon-on-glass (SOG) or silicon-on-sapphire (SOP), or epitaxial layers of semiconductor materials on another substrate. The conductivity of the substrate, or sub-regions of the substrate, may be controlled through doping using various chemical species including, but not limited to, phosphorous, boron, or arsenic. Doping may be performed during the initial formation or growth of the substrate, by ion-implantation, or by any other doping means.

(53) The functions described herein may be implemented in hardware, software executed by a processor, firmware, or any combination thereof. Other examples and implementations are within the scope of the disclosure and appended claims. Features implementing functions may also be physically located at various positions, including being distributed such that portions of functions are implemented at different physical locations.

(54) As used herein, including in the claims, “or” as used in a list of items (for example, a list of items prefaced by a phrase such as “at least one of” or “one or more of”) indicates an inclusive list such that, for example, a list of at least one of A, B, or C means A or B or C or AB or AC or BC or ABC (i.e., A and B and C). Also, as used herein, the phrase “based on” shall not be construed as a reference to a closed set of conditions. For example, an exemplary step that is described as “based on condition A” may be based on both a condition A and a condition B without departing from the scope of the present disclosure. In other words, as used herein, the phrase “based on” shall be construed in the same manner as the phrase “based at least in part on.”

(55) As used herein, the terms “top,” “bottom,” “over,” “under,” “above,” and “below” can refer to relative directions or positions of features in the semiconductor devices in view of the orientation shown in the Figures. These terms, however, should be construed broadly to include semiconductor devices having other orientations, such as inverted or inclined orientations where top/bottom, over/under, above/below, up/down, and left/right can be interchanged depending on the orientation.

(56) It should be noted that the methods described above describe possible implementations, and that the operations and the steps may be rearranged or otherwise modified and that other implementations are possible. Furthermore, embodiments from two or more of the methods may be combined.

(57) From the foregoing, it will be appreciated that specific embodiments of the invention have been described herein for purposes of illustration, but that various modifications may be made without deviating from the scope of the invention. Rather, in the foregoing description, numerous specific details are discussed to provide a thorough and enabling description for embodiments of the present technology. One skilled in the relevant art, however, will recognize that the disclosure

can be practiced without one or more of the specific details. In other instances, well-known structures or operations often associated with memory systems and devices are not shown, or are not described in detail, to avoid obscuring other aspects of the technology. In general, it should be understood that various other devices, systems, and methods in addition to those specific embodiments disclosed herein may be within the scope of the present technology.

## Claims

1. An antifuse device, comprising: a gate having a gate dielectric layer; a first doping region connected to a first end of the gate; a second doping region connected to a second end of the gate, the second end being opposite to the first end of the gate; a channel that is disposed under the gate and that connects the first doping region and the second doping region; an interconnection jumper that electrically connects the first doping region and the second doping region; and a fin that directly connects the first doping region and the second doping region, wherein the gate dielectric layer is disposed on a top surface and two sidewalls of the fin.
2. The antifuse device of claim 1, wherein a programming voltage is applied between the gate and the first doping region and/or the gate and the second doping region to break down the gate dielectric layer and short the antifuse device.
3. The antifuse device of claim 1, wherein the interconnection jumper comprises at least one of copper, tungsten, molybdenum, nickel, titanium, tantalum, platinum, silver, gold, ruthenium, iridium, rhenium, rhodium, or an alloy thereof.
4. The antifuse device of claim 1, wherein the interconnection jumper maintains the first doping region and the second doping region at a same voltage level.
5. The antifuse device of claim 4, wherein the first doping region and the second doping region are configured, through the interconnection jumper, to be grounded in order to short the antifuse device by applying a gate voltage.
6. The antifuse device of claim 1, wherein the channel is disposed under the top surface and two sidewalls of the fin.
7. The antifuse device of claim 6, wherein the gate dielectric layer is configured to break down at at least one of the top corners of the fin and bottom edges of the fin in order to short the antifuse device.
8. The antifuse device of claim 7, wherein the antifuse device is configured to be shorted by forming: a first pair of primary breakdown regions that are disposed in the channel and along the top corners of the fin respectively, the first pair of primary breakdown regions being connected to the first doping region and extending toward the second doping region, and a second pair of primary breakdown regions that are disposed in the channel and along the top corners of the fin respectively, the second pair of primary breakdown regions being connected to the second doping region and extending toward the first doping region.
9. The antifuse device of claim 8, wherein the first and second pairs of primary breakdown regions merge along the top corners of the fin.
10. The antifuse device of claim 8, wherein the gate dielectric layer adjacent to the first pair of primary breakdown regions and/or the second pair of primary breakdown regions is configured to break down in order to short the antifuse device.
11. The antifuse device of claim 1, wherein the channel, the first doping region, and the second doping region are disposed under a front surface of a substrate, and wherein the gate is disposed on the front surface of the substrate.
12. The antifuse device of claim 11, wherein the gate overhangs at least a portion of the first doping region and/or a portion of the second doping region.
13. The antifuse device of claim 12, wherein the antifuse device is configured to be shorted by forming: a first primary breakdown region disposed in the first doping region and under the gate

overhang region, and a second primary breakdown region disposed in the second doping region and under the gate overhang region.

14. The antifuse device of claim 13, wherein the gate dielectric layer adjacent to the first primary breakdown region or the second primary breakdown region is configured to break down in order to short the antifuse device.

15. An antifuse device, comprising: a fin field-effect transistor (FinFET) having a fin, a source, a drain, and a gate containing a gate dielectric layer, the source and drain of the FinFET being connected by the fin; an interconnection jumper that electrically connects the source and drain of the FinFET, wherein the gate dielectric layer is disposed on a top surface and two sidewalls of the fin, and wherein the antifuse device is configured to be shorted by forming: a first pair of primary breakdown regions that are disposed in a channel of the FinFET and along top corners of the fin respectively, the first pair of primary breakdown regions being connected to the source of the FinFET and extending toward the drain of the FinFET, and a second pair of primary breakdown regions that are disposed in the channel of the FinFET and along the top corners of the fin respectively, the second pair of primary breakdown regions being connected to the drain of the FinFET and extending toward the source of the FinFET.

16. The antifuse device of claim 15, wherein the gate dielectric layer adjacent to the first pair of primary breakdown regions and/or the second pair of primary breakdown regions is configured to break down in order to short the antifuse device.

17. The antifuse device of claim 15, wherein the gate dielectric layer is configured to break down at at least one of the top corners of the fin and bottom edges of the fin in order to short the antifuse device.

18. An antifuse device, comprising: a planar metal-oxide-semiconductor field-effect transistor (MOSFET) having a source, a drain, and a gate containing a gate dielectric layer, the gate overhangs at least a portion of the source and/or the drain; an interconnection jumper that electrically connects the source and drain of the MOSFET; and a fin that directly connects the source and the drain, wherein the gate dielectric layer is disposed on a top surface and two sidewalls of the fin, wherein the antifuse device is configured to be shorted by forming: a first primary breakdown region disposed in the source and under the gate overhang region, and a second primary breakdown region disposed in the drain and under the gate overhang region.

19. The antifuse device of claim 18, wherein the gate dielectric layer adjacent to the first primary breakdown region and/or the second primary breakdown region is configured to break down in order to short the antifuse device.

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